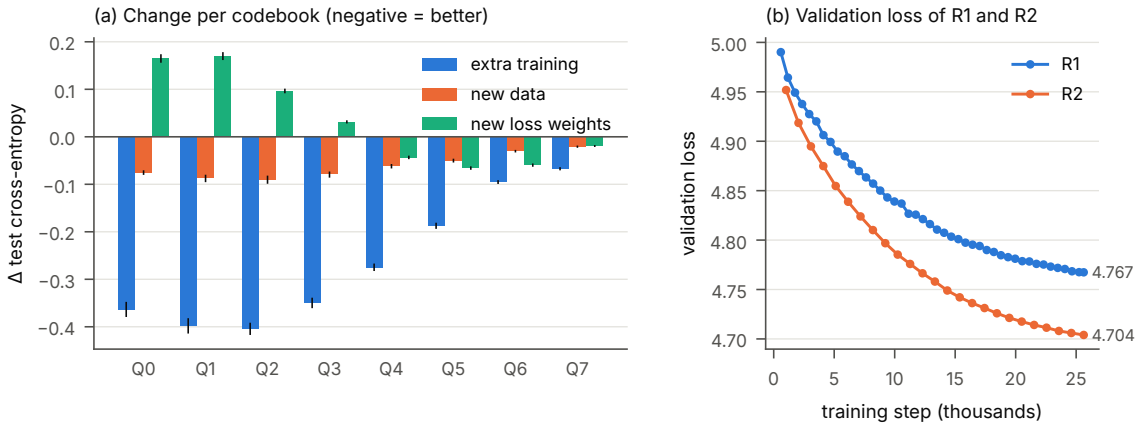
(a) Change per codebook (negative = better)
Δ test cross-entropy
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
extra training
new data
new loss weights
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
(b) Validation loss of R1 and R2
validation loss
5.00
4.95
4.90
4.85
4.80
4.75
4.70
R1
R2
4.767
4.704
0
5
10
15
20
25
training step (thousands)